

Sampling for Microbiome Studies: The Foundation of Good Science

OHMC Kickstart Workshop

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Opinion

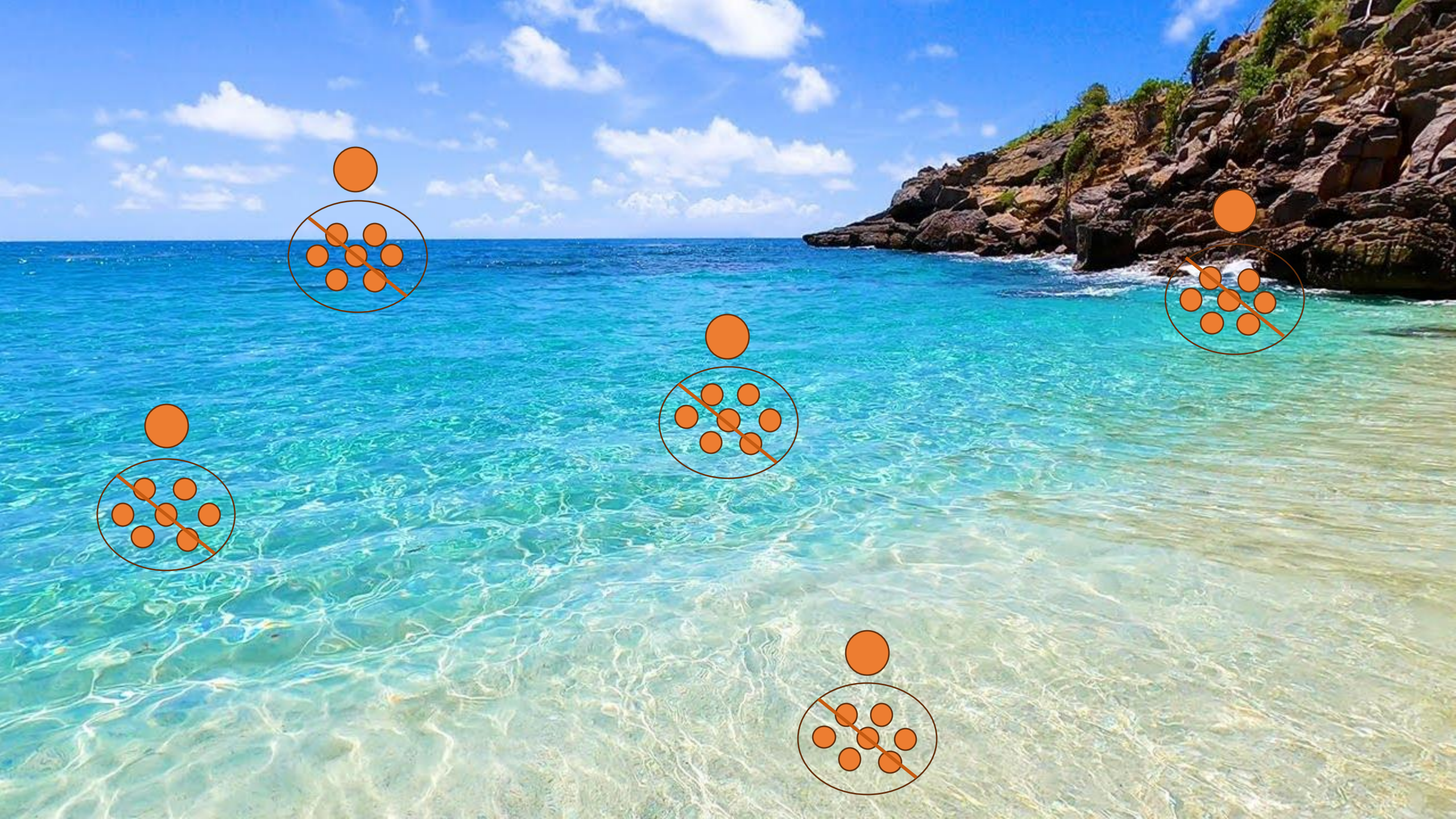
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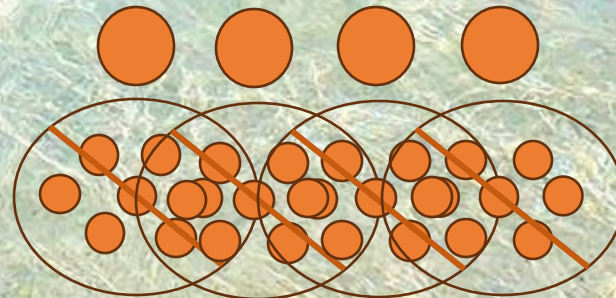
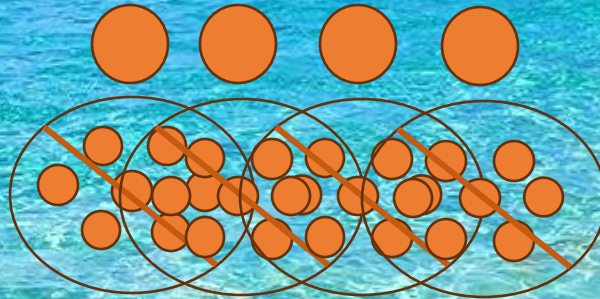
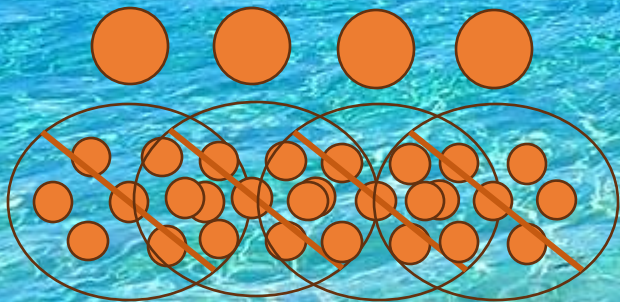
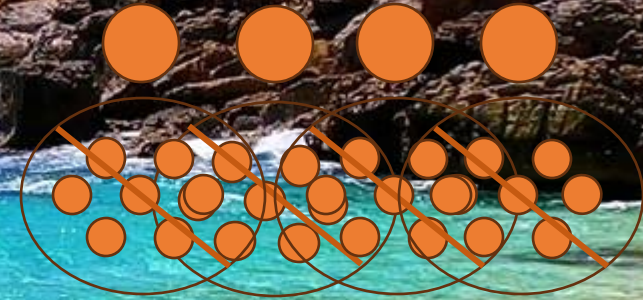
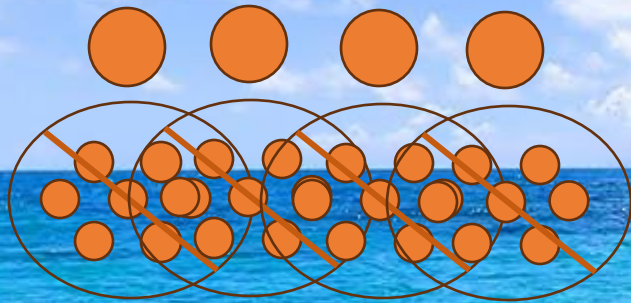
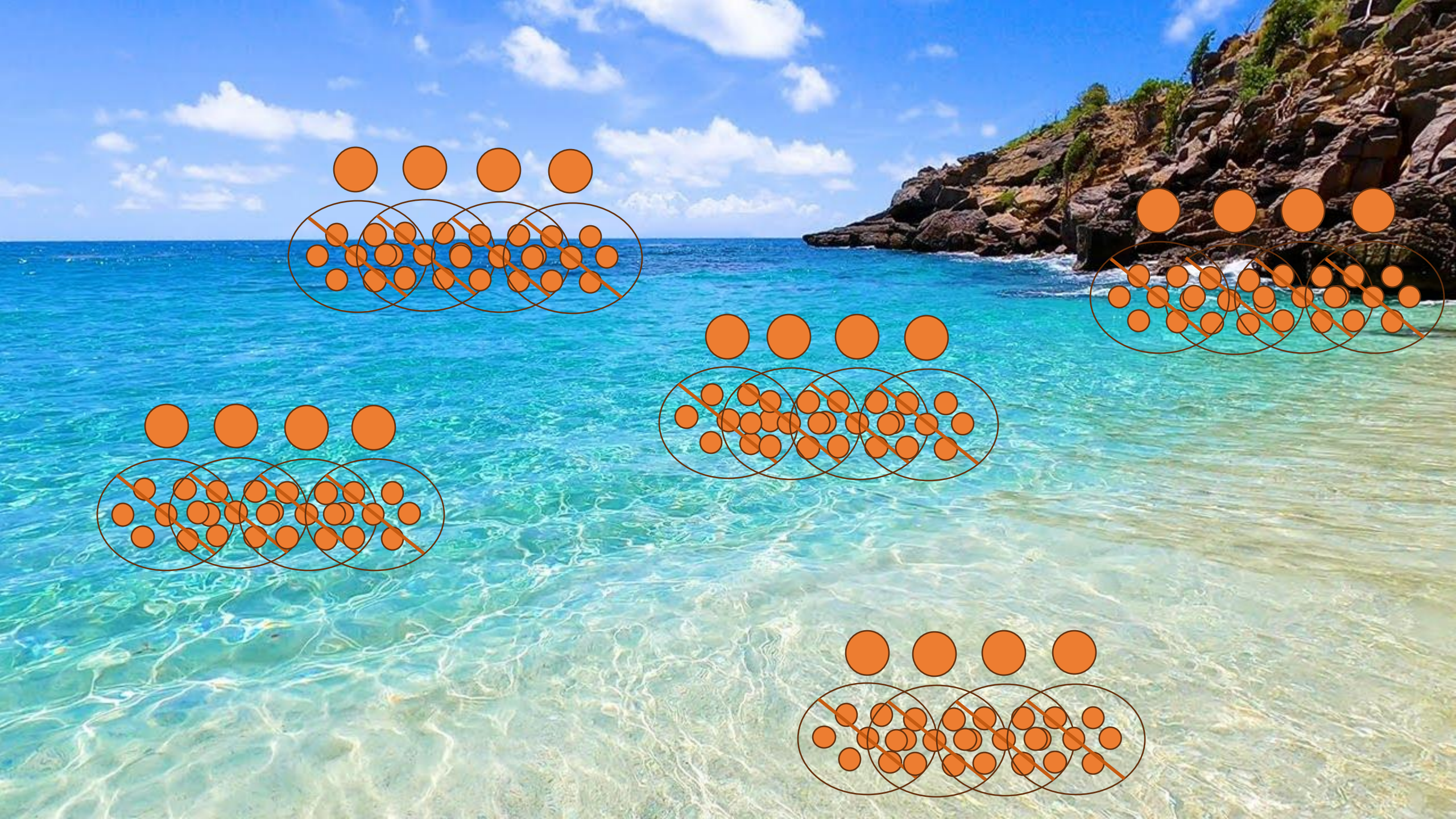
James I. Prosser*

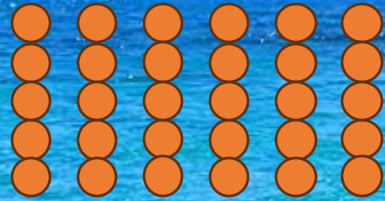
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Introduction

Andrén and colleagues (2008) recently published a paper reminding us of good scientific practice, taught and learnt during our early years of research, but frequently forgotten. This article has a similar objective. It is highly likely that the majority of microbial ecologists have taken a basic course in statistics. Unfortunately, the basic principles of statistical analysis and its significance and necessity are frequently ignored. My own exposure to this problem, and my suffering of the consequences, arise through reading, reviewing and editing articles on microbial diversity. The suffering is greatest with articles utilizing ‘expensive’ or cutting edge techniques, such as analysis of clone library sequences, microarray analysis and (increasingly) high-throughput sequencing. The problems, however, are more widespread and exist beyond these techniques and beyond studies of microbial diversity.







Temporal component

- Every hour?
- Day and night (every 12 hours)?
- Once a week?
- Every month?
- Every season?
- Once a year?

WHERE - Spatial heterogeneity

WHEN - Temporal dynamics

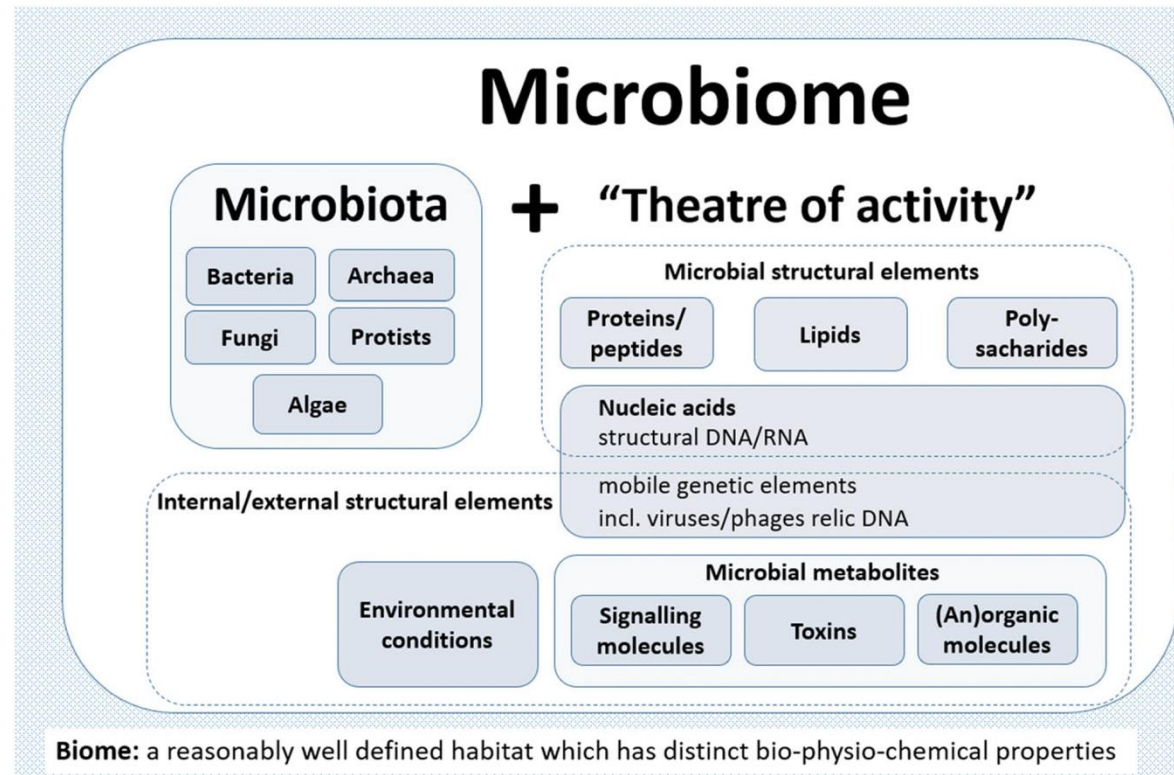
HOW - Minimize bias

Defining important terminologies

Microbiome: the microbiota + the bio-physio-chemical properties

Microbiota: the collection of microorganisms (bacteria, archaea, fungi, protists, algae)

Metagenome: the collection of all genomes (not only marker genes)



Why is sampling so critical?

The quality of our data is only as good as the quality of our samples. If your sampling is flawed, your conclusions will be flawed, no matter how sophisticated is your analysis

How to get started?



Designing a sampling strategy

Most important step

Step 1. Define your research question

Your research question is the single most important step. It's the blueprint that guides every decision you make about sampling

Examples of Bad Questions:

“What's in the sample?” (soil, gut, environmental)

“Does this sample contain a diverse microbiota?”

Why it's bad:

- Too broad, no clear objective
- Lacking mechanistic insights (why does diversity matter?)

Example of Good Question:

“What is the impact of organic farming on the soil microbiota at a depth of 10 cm as compared to a conventional system during the growing season?”

Why it's good:

Specific: Tells you what to compare (org vs. conv).
Defined: Tells you where to sample (10 cm depth).
Focused: Tells you when to sample (growing season).

Designing a sampling strategy

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Example of Good Question:

“Does the administration of a prebiotic fiber once a day lead to a measurable increase in the abundance of *Bifidobacterium* species in the human gut after a week of treatment?”

Why it's good:

Mechanistic: clear cause-and-effect relationship
Testable: controlled intervention with measurable outcome.
Specific: Focuses on a specific response

Developing a strong hypothesis – **EXTRA!!!**

Develop a deep understanding of the literature

Be specific and precise

Make it testable (and falsifiable)

Establish a clear relationship between variables

Be grounded in existing knowledge

Consider the “If-Then” format

Account for potential confounding factors



Designing a sampling strategy

Step 2. Sample size and Replication

Having the right number of samples is crucial for drawing reliable conclusions and accounting for natural variation

Sample size: how many samples do we need?

- Too few = can't trust the results (no statistical power)
- Too many = wasting time and money

Solution: Power analysis and/or know your system (literature)

Replication

- Biological Replicates: Collecting multiple samples - accounts for spatial/individual variation.
- Technical Replicates: Splitting a single sample and processing it twice – accounts for manipulation error.

Designing a sampling strategy

Step 3. Sample collection protocols

Develop a standard operating procedure (SOP)

It ensures everyone performs the steps the same way every time,
minimizing contamination and human bias

1. Use Sterile Techniques: always use sterile gloves, tools, and containers
2. Consistent Location/Method: define precisely where and how to sample
3. Consistent Time: control the time of day, season, or treatment time

Be aware of batch effects

- Changes in sampling time
- Changes in reagent lots
- Changes in personnel
- Variation in instrument performance
- Ambient conditions
- etc.

How to collect your samples? standardization and limitations

Step 3. Sample collection protocols

4. Defined Collection Device: Specify the exact tool and method + understand limitations



5. Proper Labeling: Record all essential information immediately

Designing a sampling strategy

Step 4. Storage and Transport

Proper storage and transport are critical to preserve the microbial community as it was at the moment of collection

Key preservation methods

Immediate Freezing:

- Gold Standard: -80°C freezer (prevents most microbial activity and degradation).
- Alternative: -20°C freezer (can be sufficient for short-term storage or certain applications, but less ideal).

Stabilizing Buffers:

- Use special solutions (like RNAlater or DNA/RNA Shield) that inactivate enzymes and prevent nucleic acid degradation.
Example: useful for samples that can't be frozen immediately and/or RNA based studies.

Transport

- Keep samples frozen or on ice during transit.
- Use dry ice for long-distance shipping of frozen samples.
- Use insulated coolers and ice packs for shorter distances.

The Story Behind the Sample: Standard Metadata

Key Metadata Categories

1. Sample-Specific

Unique Sample ID
Collection Date and Time
Sample Type (e.g., stool, soil, oral swab)
Collection Site & Coordinates (GPS)
Storage & Preservation Method

2. Subject/Environmental

For Human Studies: Age, Sex, BMI, Diet,
Disease State, Medication/Antibiotics,
Smoking Status.
For Environmental Studies: pH, Temperature,
Moisture, Soil Type, Land Use, Salinity.

3. Experimental & Processing

DNA Extraction Method (e.g., specific kit used)
Sequencing Platform (e.g., Illumina, PacBio)
Bioinformatics Pipeline & Software Versions, R
scripts, Date of Analysis

Minimum Information about any (x) Sequence (MlxS) standards from the
Genomic Standards Consortium ensures your data is FAIR
(Findable, Accessible, Interoperable, Reusable)

MIMARKS - Minimum Information about a Marker Gene Sequence (e.g., 16S rRNA or ITS gene studies)

MIMS - Minimum Information about a Metagenome Sequence

MlxS-Human: For samples collected from a human host

MlxS-Built Environment: For samples from human-made environments like buildings or cities

Etc.

*SAMPLING
ETHICS*



Sampling ethics

For Human Studies

- Informed Consent: Obtaining clear, voluntary permission from all participants.
- Data Privacy: Storing personal and health information securely and pseudonymizing data to protect patient privacy.
- IRB Approval: All human-subject research must be reviewed and approved by an Institutional Review Board.

For Animal Studies

- Humane Treatment: Adhering to strict standards for the care and treatment of all animals.
- IACUC Approval: All animal research must be reviewed and approved by an Institutional Animal Care and Use Committee.

For Environmental Studies

- Permits & Regulations: Obtaining permits for collecting samples from public lands, protected areas, or private property.
- Ecological Impact: Minimizing disturbance to the natural environment and local ecosystems during fieldwork.

The Ethical line: Bioprospecting *vs* Biopiracy

Sampe and processing costs

Materials for collection: sampling device, storage device, cleaning reagents, gloves, etc.

Transport and storage: dry ice, container, transport costs, etc.

Sample processing: DNA extraction, purification, PCR, library preparation, sequencing run, etc.

Analytical costs: computational costs, software, etc.



Genomics Core Facility

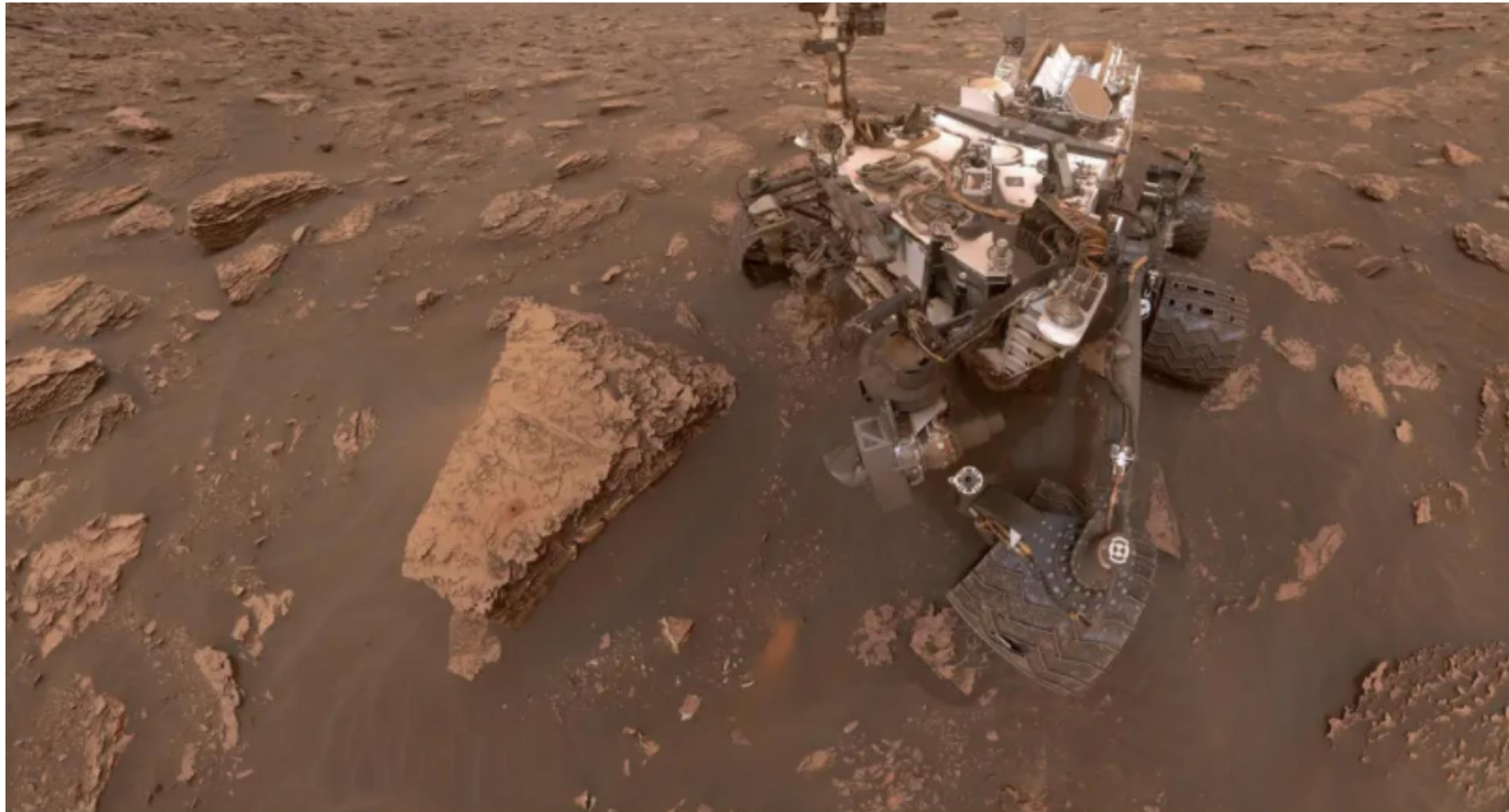
Employing next-generation sequencing and traditional genomics instruments and techniques to investigate a variety of biological phenomena



Breakout groups: Space Force Scenario!



Your Mission – Propose the best way to sample the Martian Microbiome



Sampling Scenario Priors

1. Yes, life has been discovered on Mars on a previous rover mission! Boots on ~~Moon~~ Mars!
2. Life forms are microbial and carbon based.
3. Money is NOT an issue. The sky is not the limit, space - maybe.
4. There is water under the soil surface, Mars also has a thin atmosphere with less gravity than Earth. It is considered an “extreme” environment to live in.



Sampling Teams:



Breakout Group Questions

It is your mission to propose the best way to sample the Martian microbiome. (10-15 minutes)

1. Give yourself a team name: Space Force _____
2. How will you go about sampling on Mars?

Tip: Get creative, don't get too hung up on super technical details. Design a thoughtful sampling regime.

3. What do you need to consider to make this mission successful in order to better understand the diversity, structure, and function of the Martian microbiome?
4. Extend these sampling questions to your own research project idea!



ADDITIONAL REFERENCES

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